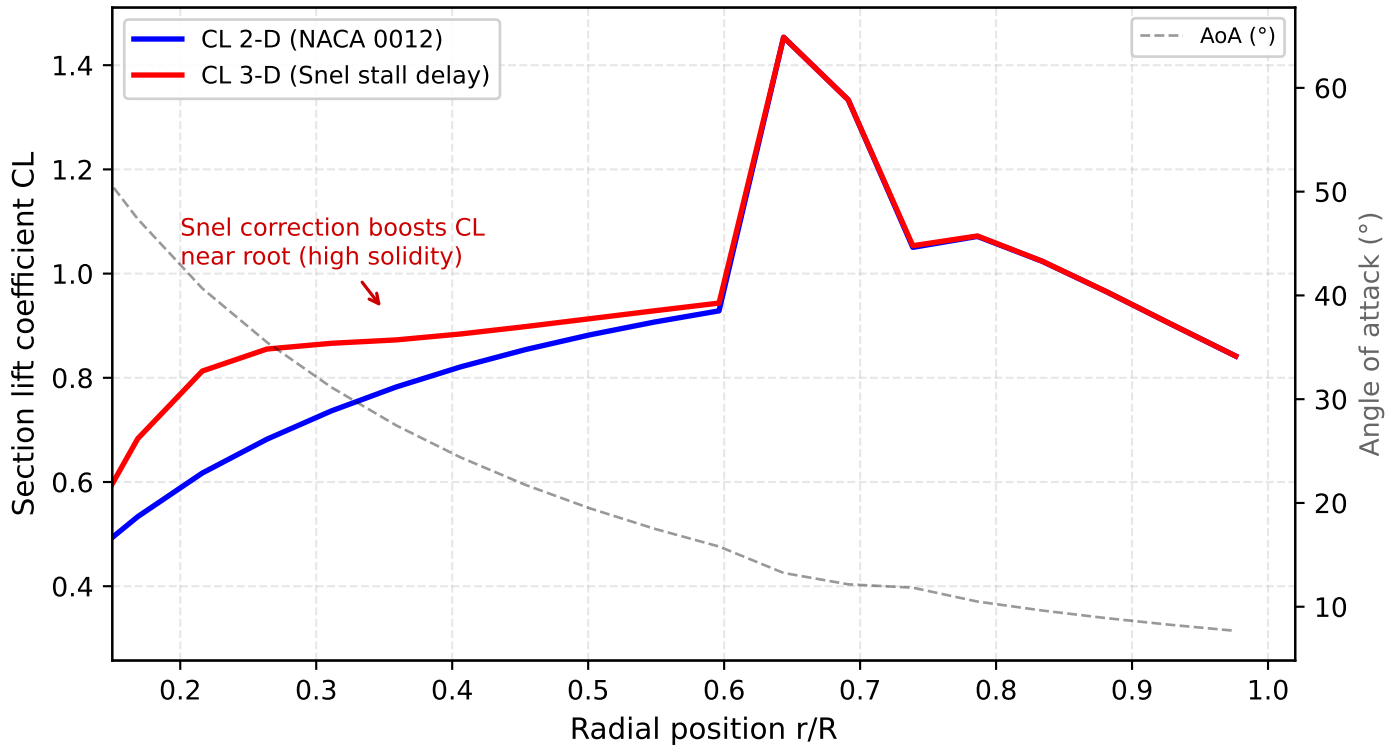


## Radial BEM Loading — R=3.0m, 8 m/s, 45° — 2-D vs 3-D Snel



*IMPLICATIONS: The Snel 3-D stall delay correction increases CL by up to 30% near the blade root ( $r/R < 0.5$ ), where the chord/radius ratio is highest and 2-D airfoil data underpredicts lift. Beyond  $r/R \approx 0.6$ , the correction vanishes — the outer blade behaves as 2-D. BEM models without stall delay systematically under-predict thrust from the inboard section. Data: R=3.0m rotor at 8 m/s wind speed, 45° elevation, 78 RPM autorotation. See SPEC.md §6.6.*